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Project Report

Personalized Risk-to-timeline modelling for cervical cancer progression:

Predicting outcomes from clinical and population risk factors.

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1.0 Abstract

Cervical cancer progression is influenced by diverse behavioral and clinical factors, yet most machine-learning research focuses only on classifying disease presence rather than modeling individualized risk trajectories. This project establishes the foundation for a personalized Risk-to-Timeline (R-to-T) framework by developing predictive models for cervical intraepithelial neoplasia (CIN) using an openly available cervical cancer risk dataset. After data cleaning and imbalance correction, several machine-learning algorithms were evaluated. Although many appeared accurate at first, cross-validation revealed the Linear SVM as the most stable and reliable model for identifying CIN, achieving strong sensitivity and balanced performance across folds. SHAP analysis further showed clinically meaningful factors central to prediction. These results provide a rigorous baseline for future time-based modelling of cervical cancer progression and support the development of transparent, personalized decision-support tools for early intervention.

2.0 Introduction

Cervical cancer remains a major global health concern, particularly in low- and middle-income countries where delayed detection and limited follow-up contribute to poorer outcomes (Ahishakiye et al. 2020). Although recent work has demonstrated strong performance of machine-learning models for early cervical cancer detection (Shakil, Islam & Akter 2024; Yang et al. 2019), most studies continue to focus on static binary classification rather than modelling individualized risk trajectories. This gap underscores the need for approaches that support personalized, time-based decision-making for cervical cancer prevention and early intervention.

Recent research shows that both traditional statistical models and machine-learning approaches can meaningfully support personalised cervical-cancer risk assessment. In a large cohort, Rothberg et al. (2018) used hierarchical regression on routine EHR data such as age, smoking, socio-economic status, HPV history; to predict CIN2+, achieving good discrimination and clearly separating women into below-, average- and above-average risk groups. Mahto and Sood (2024) demonstrated that even a small, imbalanced behavioural-risk dataset can yield strong performance when combined with ADASYN oversampling and tuned ensemble models, reporting accuracy around 97% for cervical-cancer risk prediction.

Complementing these single-study findings, Kondo, Ngondya and Rusheke (2025) showed in a systematic review that most ML studies rely heavily on the UCI cervical cancer dataset, often use incomplete preprocessing, and rarely evaluate usability or clinical deployment highlighting

important gaps around data quality, generalisability and human-centred implementation. This evidence supports our focus on personalized risk-timeline prediction while also underlining the need for preprocessing, careful handling of imbalance, and attention to implementation in real-world settings.

More recent explainable-AI approaches reinforce that key predictors such as smoking, contraception use, and STD history consistently align with known clinical risk factors, increasing model trust and interpretability (Karthikeyan et al. 2025). These findings show that data-driven tools can complement traditional screening by offering individualized risk insights.

This project extends existing cervical cancer modelling efforts by laying the groundwork for a personalized Risk-to-Timeline (R-to-T) framework. The goal is to determine whether patterns derived from population risk factors and clinical findings can support more dynamic predictions of CIN risk and inform patient-specific follow-up decisions.

The UCI Cervical Cancer Risk Factors dataset provides an appropriate starting point for this work, as it integrates a broad range of demographic, behavioral, sexual health, and diagnostic variables, including Hinselmann, Schiller, cytology and biopsy results. Given the noisy, sparse, and highly imbalanced nature of the dataset, this project sought to establish a rigorous modelling pipeline capable of generating reliable CIN predictions as a foundation for future R-to-T progression modelling.

The analysis involved extensive data cleaning and preprocessing to address missing values and standardize disparate data types, followed by strategies to reduce class imbalance and limit bias toward the majority class. A range of machine-learning algorithms was then evaluated using robust cross-validation procedures to ensure performance stability despite the small number of CIN-positive cases. From this comparison, the goal was to identify the most dependable model for CIN risk prediction while also ensuring transparency through interpretability techniques such as SHAP. Together, these steps form the methodological groundwork for advancing toward personalized, time-based risk modelling aligned with precision prevention and equitable cervical cancer care.

3.0 Methodology

3.1 Dataset Overview and Initial Inspection

The project utilized the UCI Cervical Cancer Risk Factors dataset, which contains 858 observations and 36 variables. The dataset includes key demographic characteristics,

reproductive history, behavioral risk factors, sexually transmitted disease indicators, and multiple diagnostic screening outcomes. Early inspection showed substantial inconsistencies, including a mix of numeric and textual entries, missing values, and extremely limited representation of the target outcome: only nine participants had biopsy-confirmed CIN.

3.2 Data Preprocessing and Preparation

To prepare the dataset for analysis, a comprehensive data-cleaning process was undertaken. Variables with >90 % of missing values were dropped, specifically *STDs: Time since first diagnosis* and *STDs: Time since last diagnosis*. The remaining variables were standardized by removing non-breaking spaces, aligning naming conventions, and converting all numerical fields stored as object types into appropriate numeric formats. Non-numeric placeholders, including symbols such as “?”, were reclassified as missing values. To address remaining gaps, median imputation was applied across relevant variables.

3.3 Data splitting and handling class imbalance

A critical methodological consideration in this study was the extreme imbalance in the target variable *Dx:CIN*. Only nine CIN-positive cases were present in the entire dataset (n = 858), representing just over 1% of all observations. The most considered methods to handle the imbalance were SMOTE and class weighting.

A stratified 80/20 split was applied. The training set contained only seven CIN-positive out of 686 instances and the test; 2 CIN-Positive out of 172 instances. This distribution significantly threatened the model’s ability to learn meaningful representations of CIN risk. To mitigate this imbalance, the Synthetic Minority Oversampling Technique (SMOTE) was applied to the training data to generate 10% CIN positive samples. This approach improved learnability while preserving the integrity of the test set, which remained untouched to ensure fair model evaluation. Following SMOTE augmentation, the training dataset achieved a more informative distribution of minority cases

In addition to oversampling, class weights were calculated based on inverse class frequencies and integrated into all algorithms capable of weighted learning. This ensured that misclassifying a CIN-positive sample carried a substantially higher penalty during optimisation, helping reduce majority-class dominance. By combining SMOTE with class-weighted training, the modelling pipeline established a more balanced and reliable foundation for learning CIN-related patterns.

3.4 Model Development

Nine machine learning models were trained: K-Nearest Neighbors, Multi-Layer Perceptron, Decision Tree, Gaussian Naive Bayes, Linear SVM, SVM with RBF kernel, SVM with sigmoid kernel, regularized Gradient Boosting, and Easy Ensemble. Each model was trained on the cleaned, SMOTE-augmented training dataset using standardized features where appropriate. The models were wrapped in pipelines that included a StandardScaler to normalize features where necessary.

Initial model performance was assessed using a stratified 80/20 train–test split; however, the very small number of CIN-positive cases in the untouched test set made these results unstable. To obtain a more realistic and reliable estimate of model behavior, a three-fold stratified cross-validation was then performed on the original dataset, without the SMOTE-generated samples. This approach exposed performance variations that were not visible in the single test split and provided a clearer picture of each model’s ability to identify CIN-positive cases across different subsets of the data.

3.5 Performance Evaluation

To evaluate model performance rigorously, emphasis was placed on metrics that reflect minority-class detection rather than overall accuracy. Key indicators included recall, measuring the model’s ability to correctly identify CIN-positive cases; precision, capturing the proportion of predicted positives that were truly positive; and the F1-score, which provides a balanced summary of both measures. These metrics were calculated specifically for the CIN-positive class to reflect the clinical importance of identifying true cases. During cross-validation, scores were averaged across folds to obtain a more stable and generalizable assessment of each model’s predictive performance.

3.6 Feature Importance and Explainability

To enhance interpretability, SHAP analysis was performed on the final Linear SVM model trained on the full dataset. SHAP values quantify the marginal contribution of each feature to a specific prediction. KernelExplainer was used to generate global and local explanations, providing insight into how specific risk factors influenced CIN predictions and enabling alignment with known epidemiological pathways.

4.0 Results

Initial model evaluation using a stratified 80/20 train–test split (Table 1) suggested exceptionally high performance across nearly all algorithms; however, these results were heavily influenced

by the very small number of CIN-positive cases in the untouched test set. Models such as the Decision Tree, Linear SVM and Gradient Boosting appeared to achieve perfect accuracy and F1-scores, yet these metrics were based on only two positive samples and therefore lacked reliability. Even models that performed less convincingly in later analysis, such as GaussianNB, KNN and the RBF SVM recorded accuracy values above 99%, illustrating that accuracy alone was not an informative measure under such extreme class imbalance.

A more reliable assessment was obtained through a three-fold stratified cross-validation applied to the original dataset without SMOTE-generated samples (Table 2). Cross-validation exposed meaningful performance differences that were not apparent in the single split. The Linear SVM emerged as the most robust and balanced model, achieving perfect recall (1.00) and high precision (0.917), resulting in the strongest F1-score (0.952) for CIN-positive predictions. The RBF SVM also performed well but showed reduced sensitivity, while the Decision Tree demonstrated high recall at the cost of precision.

Other models, including Gradient Boosting, GaussianNB, the MLP, and the Sigmoid SVM, struggled to consistently detect minority-class cases, reflected in lower recall and F1-scores. The Easy Ensemble produced high recall but very poor precision, making it prone to false positives and therefore unsuitable for risk prediction where specificity is important.

Table 1. Single 80-20 split with Weighted Metrics

Models	Accuracy	Precision	Recall	F-1 Score
Decision Trees	100%	100%	100%	100%
SVM (Linear)	100%	100%	100%	100%
Gradient Boosting(Regularized)	100%	100%	100%	100%
KNN	99.4%	99.4%	99.4%	99.3%
GaussianNB	99.4%	99.4%	99.4%	99.3%
SVM (RBF)	99.4%	99.4%	99.4%	99.3%
Easy Ensemble	98.3%	99.3%	98.3%	98.6%
SVM(Sigmoid)	97.7%	99.2%	97.7%	98.2%
MLP	98.8%	97.7%	98.8%	98.3%

Table 2. 3-Fold Stratified CV Metrics (Mean)

Models	Accuracy	Precision	Recall	F-1 Score
SVM (Linear)	99.9%	91.7%	100%	95.2%
SVM (RBF)	99.9%	100%	88.9%	93.3%
Decision Trees)	99.8%	83.3%	100%	90.5%
GaussianNB	99.8%	100%	77.8%	83.3%
KNN	99.5%	100%	55.6%	66.7%
Gradient Boosting(Regularized	99.4%	58.3%	55.6%	55.2%
Easy Ensemble	98.3%	40%	100%	56.5%
MLP	96.5%	10.4%	33.3%	15.6%
SVM(Sigmoid)	90.8%	19.3%	100%	29.3%

Together, Tables 1 and 2 show that cross-validation offers a more realistic evaluation of model behavior under severe class imbalance. While the train–test split obscured key differences, cross-validation revealed that the Linear SVM consistently achieved the best balance of sensitivity and precision for CIN detection. This makes it the most reliable model for identifying CIN-positive cases and the strongest candidate for advancing into the Risk-to-Timeline modelling framework.

In addition to quantitative model evaluation, SHAP analysis was used to examine how the best-performing model; the Linear SVM arrived at its predictions. The SHAP summary plots indicated that clinically meaningful variables, such as HPV status, prior cancer or abnormal diagnostic findings, and selected behavioral risk factors, contributed most strongly to CIN-positive predictions. This alignment with known clinical risk patterns provides further support for the model’s interpretability and strengthens confidence in its suitability for integration into the Risk-to-Timeline framework.

5.0 Discussion

The findings of this study show how challenging it can be to develop reliable machine-learning models for CIN prediction when working with a small and highly imbalanced dataset. The initial train–test split (Table 1) appeared to show near-perfect model performance, but these results were largely driven by the very small number of CIN-positive cases in the test set. With only two positive samples available, accuracy became an unreliable measure and masked important weaknesses across several models.

The cross-validation analysis (Table 2) provided a more realistic picture of how the models behave when repeatedly exposed to different subsets of the data. The Linear SVM consistently offered the best balance between identifying true CIN-positive cases and avoiding false positives, making it the most dependable option under the dataset's severe class imbalance. Other models such as the RBF SVM and Decision Tree, showed strengths but also notable trade-offs, either missing some positive cases or over-predicting them. More complex models, including the MLP and Easy Ensemble, struggled to detect CIN-positive samples with any consistency.

The results indicate that the Linear SVM provides the most dependable foundation for the next steps of the Risk-to-Timeline modelling framework. Its strong minority-class performance, combined with its interpretability enhanced further through SHAP explanations, makes it well suited for integration into a clinical decision-support pipeline where transparency is essential. The model's reliance on clinically meaningful variables, such as HPV status and abnormal diagnostic tests, further supports its potential applicability in real-world settings.

However, the limitations of the dataset must also be acknowledged. The absence of temporal information prevents direct modelling of disease progression, and the small number of CIN-positive cases limits generalizability. SMOTE improved training behavior but cannot substitute for genuine clinical diversity.

6.0 Conclusion

This study set out to build a reliable predictive foundation for personalized Risk-to-Timeline modelling using the UCI Cervical Cancer Risk Factors dataset. Despite the poor dataset quality, the modelling pipeline successfully identified the Linear SVM as the most dependable algorithm for CIN prediction. Cross-validation demonstrated that this model achieved the best balance of sensitivity and precision, and SHAP analysis confirmed that its predictions were driven by clinically meaningful features such as HPV status and diagnostic test outcomes. These results highlight both the potential of machine-learning approaches in supporting early cervical cancer risk assessment and the importance of rigorous evaluation methods when working with small, imbalanced clinical datasets. Overall, the findings provide a solid technical and conceptual foundation for developing more advanced, time-based risk models that could support earlier intervention and more personalized follow-up strategies.

6.1 Dataset Limitations

There is a lack of longitudinal data for individual patients. The dataset contains a once-off cross-sectional collection and captures disease prevalence unlike incidences over time. The original dataset is also highly imbalanced. Despite applying strategies like SMOTE and class weighting, the small percentage of positive cases makes it more difficult for the models to accurately focus on personalized prediction.

7.0 Future Research Considerations

Future work will address the dataset's limitations and advance the modelling framework toward full Risk-to-Timeline prediction. Expanding the dataset, particularly with more CIN-positive and longitudinal records will be essential for improving model generalizability and enabling time-to-event analysis. Methods such as survival modelling or time-aware machine-learning approaches may be explored to better capture progression patterns. External validation and model calibration will also be necessary to ensure clinical reliability. Ultimately, refining and integrating the model into a transparent decision-support tool could enhance screening decisions and support more personalized follow-up strategies.

8.0 References

- Ahishakiye, E., Wario, R.D., Mwangi, W. and Taremwa, D., 2020. Prediction of Cervical Cancer Basing on Risk Factors using Ensemble Learning. In: IST-Africa 2020 Conference Proceedings. IST-Africa Institute and IIMC. Available at: <https://www.researchgate.net/publication/341204214>
- Karthikeyan, P., Chandrakala, A., Shankar, T., Elangovan, D. & Ravi, V. (2025) 'Explainable AI based cervical cancer prediction using demographic, behavioural and clinical data', Scientific Reports. <https://www.nature.com/articles/s41598-025-23593-9>
- Kondo, T.S., Ngondya, D. & Rusheke, H. (2025) 'Application of artificial intelligence in cervical cancer diagnosis using risk factors: A systematic review', Telematics and Informatics Reports, 20, 100250.
<https://www.sciencedirect.com/science/article/pii/S2772503025000647#:~:text=The%20UCI%20dataset%20has%20been%20extensively%20used,to%20a%20varying%20degree%20by%20different%20researchers>.
- Mahto, R. & Sood, K. (2024) 'Predicting Cervical Cancer Based on Behavioral Risk Factors', International Journal of Advanced Computer Science and Applications, 15(11).
https://thesai.org/Downloads/Volume15No11/Paper_1-Predicting_Cervical_Cancer_Based_on_Behavioral_Risk_Factors.pdf
- Rothberg, M.B., Haessler, S.D., Lagu, T., Lindenauer, P.K. & Pekow, P.S. (2018) 'A risk prediction model to allow personalized screening for cervical cancer', Cancer Causes & Control, 29(3), pp. 297–304
<https://pubmed.ncbi.nlm.nih.gov/29450667/#:~:text=Results:%20The%20final%20dataset%20included.more%20targeted%20cervical%20cancer%20screening>.
- Shakil, R., Islam, S. and Akter, B., 2024. A precise machine learning model: Detecting cervical cancer using feature selection and explainable AI. Journal of Pathology Informatics, 15, p.100398. <https://doi.org/10.1016/j.jpi.2024.100398> .
- Yang, W., Gou, X., Xu, T., Yi, X. and Jiang, M., 2019. Cervical Cancer Risk Prediction Model and Analysis of Risk Factors based on Machine Learning. In: Proceedings of the 2019 International Conference on Bioinformatics and Biomedical Technology (ICBBT). ACM, pp.50-55.
<https://doi.org/10.1145/3340074.3340078> .